Module 1: dplyr Practice

Lesson 3 | Work with the Restaurant CSV

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Preparation

Read Lesson 3. Load `data/module02/restaurant_ratings.csv` from the course project. Preserve the original values and ordered quality categories. Use pipes and save each requested result under a descriptive name.

Tasks

1. Inspect column types, row count, missing values, and duplicate restaurant identifiers. State what one row means.
2. Select restaurant ID, quality, and price. Retain Excellent restaurants priced at least USD 30. How many rows remain?
3. Return the five highest-priced restaurants, ordering tied prices by restaurant ID.
4. Create four price bands [10,20), [20,30), [30,40), [40,50), and a logical indicator for price at least USD 30. Verify that no price is left unclassified.
5. Count restaurants by quality category. Calculate group size, mean price, and median price using `group_by()` and `summarise()`. Return an ungrouped result.
6. Create a lookup containing one median price per quality group, then left-join it to the restaurant rows. Verify the row count and explain why duplicate lookup keys could change it.
7. Reshape the mean and median summary columns into a long table with `measure` and `price_usd` columns. Reshape it back to wide form.
8. Write a brief explanation of why repeated prices are not necessarily duplicate observations, and why ignoring missingness can mislead an analysis.

Return to Module 1.